

Week 11 Bayesian Decision Theory

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Duda, Hart and Stork, Dr. Gavin Brown

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Bayes Decision Theory

- Statistically best classifier
- Uses a probabilistic approach
- The theory assumes:
 - *Decision problem can be posed in probabilistic terms*
 - *All relevant probability values are known (in practice this is not true)*

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Introduction

- The sea bass/salmon example
 - *State of nature*
 - *a priori (prior) probability*
 - *State of nature is unpredictable, so it is a random variable*
 - *The catch of salmon and sea bass is equiprobable*
 - $P(\omega_1) = P(\omega_2)$ (uniform priors)
 - $P(\omega_1) + P(\omega_2) = 1$
 - *Prior prob. reflects our prior knowledge about how likely we are to observe a sea bass or salmon*
 - *These probabilities may depend on time of the year or the fishing area!*

Decision Rule

- Decision rule with only the prior information
 - *Decide ω_1 if $P(\omega_1) > P(\omega_2)$, otherwise decide ω_2*
- Suppose now we have a measurement or feature on the state of nature - say the fish lightness value
- Use of the class –conditional probability density $P(x | \omega_1)$ and $P(x | \omega_2)$ describe the difference in lightness feature between populations of sea bass and salmon

Class Conditional Probabilities

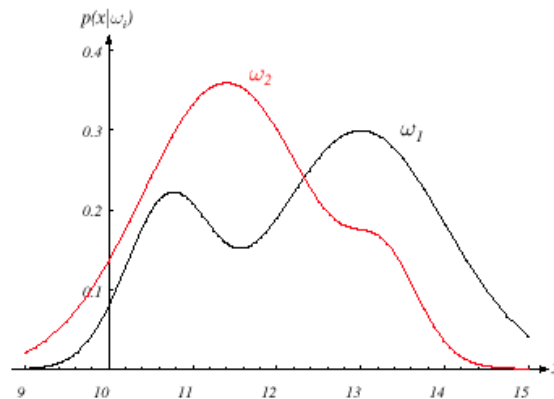


FIGURE 2.1. Hypothetical class-conditional probability density functions show the probability density of measuring a particular feature value x given the pattern is in category ω_i . If x represents the lightness of a fish, the two curves might describe the difference in lightness of populations of two types of fish. Density functions are normalized, and thus the area under each curve is 1.0. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons,

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Class Conditional Probabilities

- Decision rule
 - Assign input pattern x to class ω_1 if
$$P(x | \omega_1) > P(x | \omega_2), \text{ otherwise } \omega_2$$
- Suppose, we know prior and conditional probabilities and that we have observed the value of the feature x
 - How would you decide on the “state of nature”—type of fish, based on this info?
 - Bayes theory allows us to compute the posterior probabilities from prior and class-conditional probabilities

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Posterior Probability

Posterior, likelihood, evidence

- Using Bayes formula

$$P(w_j|x) = \frac{P(x|w_j) P(\omega_j)}{P(x)}$$

where in case of two categories

$$P(x) = \sum_{j=1}^2 P(x|\omega_j)P(\omega_j)$$

- Posterior = (Likelihood $\times$ Prior) / Evidence
- Evidence $P(x)$ can be viewed as a scale factor that guarantees that the posterior probabilities sum to 1
- $P(x|\omega_j)$ is called the likelihood of ω_j with respect to x ; the category ω_j for which $P(x|\omega_j)$ is large is more likely to be the true category

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Posterior Probability

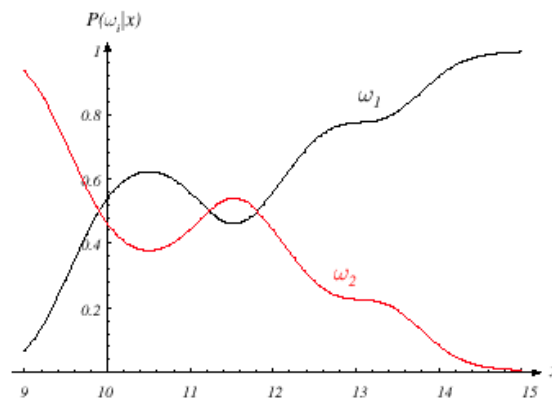


FIGURE 2.2. Posterior probabilities for the particular priors $P(\omega_1) = 2/3$ and $P(\omega_2) = 1/3$ for the class-conditional probability densities shown in Fig. 2.1. Thus in this case, given that a pattern is measured to have feature value $x = 14$, the probability it is in category ω_2 is roughly 0.08, and that it is in ω_1 is 0.92. At every x , the posteriors sum to 1.0. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

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Bayes Decision Rule

- $P(\omega_1 | x)$ is the probability of the state of nature being ω_1 given that feature value x has been observed
- Decision given the posterior probabilities, [Bayes Decision rule](#)

x is an observation for which:

if $P(\omega_1 | x) > P(\omega_2 | x)$  True state of nature = ω_1

if $P(\omega_1 | x) < P(\omega_2 | x)$  True state of nature = ω_2

That is Choose the class that has the larger posterior probability

Probability of Error

- Therefore, whenever we observe a particular x , the probability of error is:

$$P(\text{error} | x) = P(\omega_1 | x) \text{ if we decide } \omega_2$$

$$P(\text{error} | x) = P(\omega_2 | x) \text{ if we decide } \omega_1$$

- Bayes decision rule minimizes the probability of error
- Decide ω_1 if $P(\omega_1 | x) > P(\omega_2 | x)$; otherwise decide ω_2

Therefore:

$$P(\text{error} | x) = \min [P(\omega_1 | x), P(\omega_2 | x)]$$

Bayes Decision Rule

Optimal Bayes decision rule

Decide ω_1 if $P(\omega_1 | x) > P(\omega_2 | x)$; otherwise decide ω_2

Special cases:

(i) $P(\omega_1) = P(\omega_2)$; Decide ω_1 if
 $P(x | \omega_1) > P(x | \omega_2)$, otherwise ω_2

(ii) $P(x | \omega_1) = P(x | \omega_2)$; Decide ω_1 if
 $P(\omega_1) > P(\omega_2)$, otherwise ω_2

The Normal (Gaussian) Density

- Univariate density: $N(\mu, \sigma^2)$
 - Continuous univariate normal or Gaussian density

$$P(x | w_j) = \frac{1}{\sqrt{2\pi} \sigma_j} \exp \left[-\frac{1}{2} \left(\frac{x - \mu_j}{\sigma_j} \right)^2 \right]$$

where:

μ = mean (or expected value) of x

σ^2 = variance (or expected squared deviation) of x

- Mean or expected value and variance of x is given by

$$\mu = \sum_{k=1}^{k=n} \frac{x_k}{n} \quad ; \quad \sigma^2 = \frac{\sum_{k=1}^{k=n} (x_k - \mu)^2}{n}$$

$$E(x) = \mu = \sum_{x \in X} x \cdot P(x) = \sum_{i=1}^m x_i \cdot p_i$$

$$Var[x] = \sigma^2 = E((x - \mu)^2) = \sum_{x \in X} (x - \mu)^2 \cdot P(x)$$

The Normal Density

- The Gaussian distribution with zero mean and unit variance, denoted as $N(0,1)$, is called *standard normal distribution*.

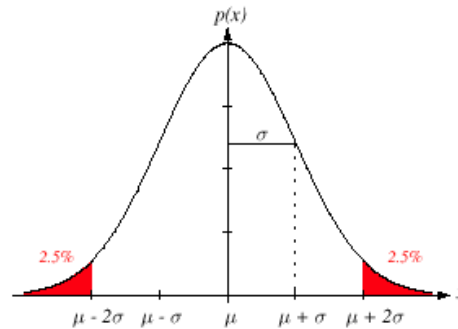


FIGURE 2.7. A univariate normal distribution has roughly 95% of its area in the range $|x - \mu| \leq 2\sigma$, as shown. The peak of the distribution has value $p(\mu) = 1/\sqrt{2\pi}\sigma$. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

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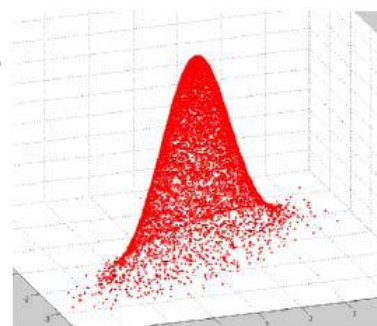
Multivariate Density: $N(\mu, \Sigma)$

Multivariate normal density in d dimensions:

$$P(x | w_j) = \frac{1}{(2\pi)^{d/2} |\Sigma_j|^{1/2}} \exp \left[-\frac{1}{2} (x - \mu_j)^t \Sigma_j^{-1} (x - \mu_j) \right]$$

where:

$x = (x_1, x_2, \dots, x_d)^t$
 $\mu = (\mu_1, \mu_2, \dots, \mu_d)^t$ mean vector
 $\Sigma = d \times d$ covariance matrix
 $|\Sigma|$ and Σ^{-1} are determinant and inverse of Σ , respectively



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Covariance matrix Σ

- Covariance defines how the variables vary together as a pair, are they both increasing together, does one increase when the other decrease, etc.
- The diagonal elements σ_{ii} are the variances of the respective x_i (i.e., σ_i^2), and the off-diagonal elements σ_{ij} are the covariances of x_i and covariance x_j

$$\hat{\Sigma} = \frac{\sum_{k=1}^n (x_k - \hat{\mu})(x_k - \hat{\mu})^t}{n}$$

$$\Sigma = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{bmatrix}$$

- We would expect a positive covariance for the length and weight features of a population of fish
- If x_1 and x_2 are statistically independent then the covariance of x_1 and x_2 is zero i.e., $\sigma_{ij} = 0$
- The covariance matrix becomes a diagonal matrix, with the individual variances in the main diagonal

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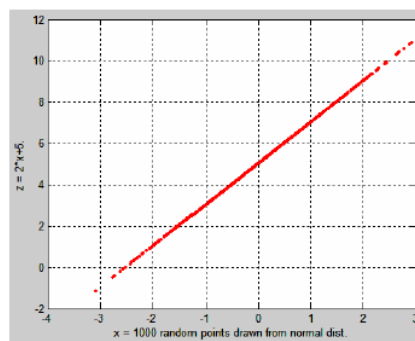
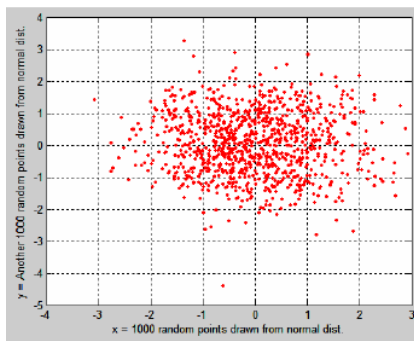
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Co-variance Matrix Properties: Example



$$\Sigma_{xy} = \begin{bmatrix} 1.073 & -0.026 \\ -0.0264 & 0.9673 \end{bmatrix}$$

$$\Sigma_{xz} = \begin{bmatrix} 1.073 & 2.1476 \\ 2.1476 & 4.2951 \end{bmatrix}$$

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